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Convolutional Vision Transformer for Handwritten Digit Recognition

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Abstract

Handwritten digit recognition is an essential step in understanding handwritten documents. The state-of-the-art convolutional neural networks (CNN) methods are mostly preferred for recognizing handwritten digits. Though the recognition accuracy is high, CNN filter weights don't change even after training. Hence the process cannot adjust dynamically to changes in input. Recently the interest of researchers of computer vision has been on Vision Transformers (ViTs) and Multi-layer Perceptrons (MLPs). The use of transformer architecture enabled substantial parallelization and translation quality improvement. The inadequacies of CNNs sparked a hybrid model revolution, which combines the best of both disciplines. This paper is written to view the impact of the hybrid model on handwritten digit recognition. The testing is done on the available benchmark datasets, the Extended Modified National institute of standards and technology (EMNIST) digits dataset, and the most significant historical handwritten digit dataset (DIDA). The 10-fold cross-validation accuracy achieved on EMNIST and DIDA is 99.89% and 99.73%, respectively. The results show that the proposed method achieves the highest accuracy compared to existing methodologies. The proposed method is robust, feasible, and effective on clean and uncleaned images.

Keywords: Convolutional Neural Network, Vision Transformer, Handwritten Digit Recognition, Machine Learning, Computer Vision

1 Introduction

Handwritten digit recognition (HDR) is considered an essential task in the optical character recognition field (1). But the variations in handwriting styles made HDR challenging. Though lots of research on HDR resulted in the proposal of different techniques, an effective and robust method is still required for uncleaned and noised images. The critical element for the improvement of robustness in vision networks is attention. A

deep learning algorithm that incorporates attention processes to weigh the importance of elements of the input images differently is known as a transformer in machine learning (ML). Attention is one of the necessary blocks of ML transformers (50; 17). They are made up of many self-attention layers. The use of transformers has increased in tasks related to natural language processing (NLP) (51) and computer vision (CV) (13). They hold great potential for a general learning approach that can be used for various data

sources, especially recent advancements in computer vision that have achieved a high accuracy rate while reducing resource usage.

Currently, CNN is the state-of-the-art method for tasks related to computer vision. Attention is employed in computer vision either in combination with CNN or to replace some parts of CNN while maintaining their overall composition. Recently the interest of researchers increased in Vision transformers (ViTs) (26) and Multilayer perceptrons (MLPs) (47). The architecture of ViT is adapted from transformer architecture (17). A. Dosovitskiy et al. (26) introduced ViT in 2021. ViT models are outperforming CNN in terms of efficiency and accuracy. But the performance of ViT degrades for smaller datasets, and CNN performs better on it. The difference between ViT and CNN is as follows.

1. ViT uses fewer resources as compared to CNN.
2. CNN achieves better accuracy while training smaller datasets, whereas ViT relies on data augmentation and regularization.
3. ViT tokenizes images, and CNN employs pixel arrays.
4. ViT captures more relevant data than ResNet at lower levelsNet, resulting in substantially different features (38).
5. While targeting a random region of interest, ViT models outperform CNN (16).

The paper is organized as follows. Section 2 explores the latest techniques used for HDR. The vision transformer's detailed architecture and mathematical formulae are explained in section 3. Section 4 provides a glimpse of the proposed approach. The details of the datasets used for performance evaluation and the results are shown in section 5. It also explores the hyper-parameter optimization technique used to decide model parameters' value. Finally, section 6 concludes the paper.

2 Related Work

Various technologies have been used for HDR, and still, the research is on. As per the Scopus database journal articles, Deep Digit Recognition Network (DDRNet) based on Deep ConvNets (46) is the latest CNN model used for HDR. It reduces the number of parameters and features

used but increases accuracy with adaptive voting (AV) scheme. The model originated from the You Only Look Once (YOLO) (39) algorithm. T. Senthil et al. (42) modified the network structure of CNN with Orthogonal Learning Particle Swarm Optimization (CNN-OLPSO). This update is used to optimize the number of hyper-parameters over time. They also modeled Layered Convolutional Neural Networks (LCNN) with Squirrel Optimizer (LCNN-SO) (41). The proposed model added multiple layers to CNN to improve classification accuracy.

X. He et al. (21) combined deep neural network (DNN) with ghost imaging (GI) for HDR. Here features are extracted from the signals generated by cosine transform speckle. K. Urazoe et al. (49) used the combination of CNN with its learning method for the rotation invariant HDR. As per their experimental results, ResNet-50 was the best combination.

S. Ali et al. (6) incorporated CNN with an extreme learning machine (ELM) (31) algorithm and rectified linear unit (ReLU) activation function. They developed a deeplearning4j (DL4J) framework-based CNN-ELM model. Z. Shaukat et al. (43; 5) recognized handwritten digits using a cloud-based neural network and DL4J framework. Likewise authors used various neural network-based models for HDR CNN with parameters modification (4; 55; 2), adaptive fractional-order back-propagation neural network (14), deep convolutional self-organizing maps (DCSOM) network (7), scalable neural network (25), CNN with particle swarm optimization (PSO) (10), CNN based on the HfO₂ memristor neuron (52), adaptive Q-learning deep belief network (37), spiking neural networks (27), multilayer perceptron neural network (30).

T. A. Assegie and P. S. Nair (8) recognized handwritten digits using the decision tree classification approach of machine learning. Y. Wang et al. (53) proposed using the quantum k-nearest neighbor algorithm for HDR. Here Grover algorithm was used to find the same k points. The dimensionality reduction techniques, principal component analysis (PCA), and linear discriminant analysis (LDA) (19) are used by R. Sheikh and M. Patel (45) for HDR. They found that for smaller datasets, the performance of PCA is better than LDA. Consensus Clustering (CC)

(32) is another type of machine learning technique used for HDR.

As per (11; 33; 35) ViT is better for larger datasets and improves performance and robustness. CNN is better for smaller datasets, and the architecture is capable of extracting features. It is invariant to shift and scale. Hence this paper proposes using a combination of CNN and ViT for the handwritten digit recognition(HDR) task of computer vision. The performance of the proposed approach is evaluated on cleaned, denoised digit images as well as on images with original appearance.

3 Vision Transformer Architecture

The Fig. 1 shows the architecture of the vision transformer (26). Embedded patches are given as input to the transformer. The image is first divided into the fixed-size patch to generate the embedding. Padding is applied to overcome the difference between image size and patch size. Then the patches are projected into a 1D vector, as the transformer takes a 1D sequence as input. Positional embedding is also added to learn the positional information of each patch. The transformer encoder contains a multi-head attention layer and an MLP layer, also known as a feed-forward network layer. Layer normalization (9) is applied to both the layers individually. For 1D sequence vector v , the formulae for layer normalization is given in equation (1).

$$(v) = \gamma \frac{v - \mu}{\sigma} + \beta \quad (1)$$

where,

μ is the mean of elements in v

σ is the standard deviation of elements in v

γ is the scale parameter, and

β is a biased vector parameter

The mechanism used in multi-head attention is self-attention. The input given to self-attention is a query (Q), key (K), and value (V). It maps Q and K-V pairs to the output. Here the output is the weighted sum of the value vector, where weight is computed using the softmax function. Attention

is defined as shown in equation (2) (40).

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (2)$$

where, d is the hidden dimensions. The multi-head layer divides the input into small parts and computes the attention individually and parallelly.

The MLP layer is a two-layer feed-forward network with a gaussian error linear unit (GeLU) activation function (22). GeLU is defined as in equation (3) (22)

$$GeLU(x) = x\Phi(x) \quad (3)$$

$$\approx 0.5x(1 + \tanh[\sqrt{2/\pi}(x + 0.044715x^3)])$$

where, $\Phi(x)$ is the Gaussian cumulative distribution

4 Proposed Approach for Handwritten Digit Recognition

The recent datasets of handwritten digits are enormous as the Extended Modified National institute of standards and technology (EMNIST) digits dataset (15) contains 280,000 images, and the largest historical handwritten digit dataset (DIDA) (29; 28) with 250,000 images. This paper uses a combination of CNN and ViT, a convolutional vision transformer (CVT) (54), to recognize handwritten digits.

The dataset images are initially resized so that all the images are of the same size when sent to the model. It increases the model performance. Then the images are normalized using mean and standard deviation. Normalization is also known as rescaling and helps apply the same algorithm to all images. Scaling all images contributes to total loss and provides a standard learning rate.

The architecture of the convolutional vision transformer is presented by H. Wu et al. (54). The architecture is divided into three stages. Each stage consists of a CNN, layer normalization, and vision transformer. For the implementation of handwritten digit datasets, we modified the parameters of the transformer. Especially the value of the depth parameter is changed to improve model performance and accuracy. The

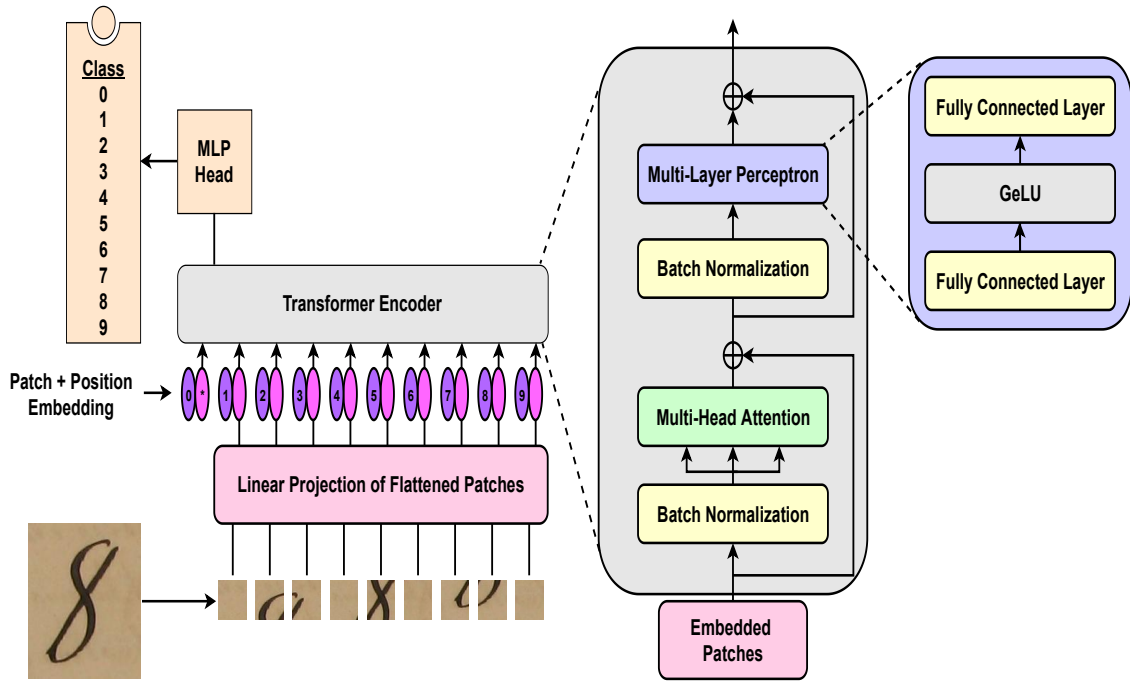


Fig. 1 Architecture of Vision Transformer

CNN is primarily used for the computation of embedding patches and feature extraction.

5 Experimental Setup

5.1 Datasets

The two main benchmark datasets used for experimentation are EMNIST-digit and DIDA. The EMNIST-digit dataset consists of 280,000 images, of which 252,000 images are used for training and 28,000 for testing. As cross-validation is applied, training and testing images change after every fold. The dataset is divided into ten balanced classes 0,1,2,3,4,5,6,7,8,9. Fig. 2 shows sample image of this dataset.

The images in the EMNIST-digit dataset are size-normalized, denoised, and cleaned. That's why another benchmark dataset DIDA is used to check the effectiveness and robustness of the proposed model. The images of the DIDA dataset are in original size and appearance. The DIDA dataset consists of 252,860 images cropped from the historical documents, of which 227,574 images are used for training, and 25,286 images are used for testing. The digits in the DIDA dataset appear with different writing styles, sizes, directions,

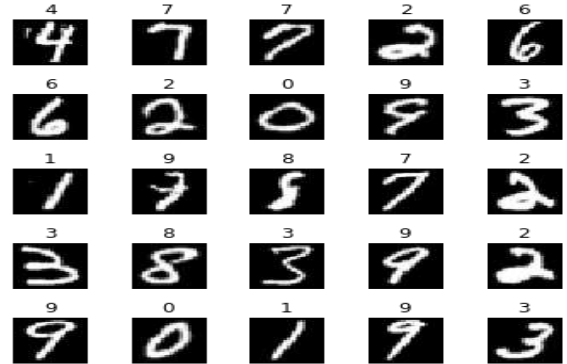


Fig. 2 Samples of digits in EMNIST dataset

widths, and arrangements. The DIDA dataset is also divided into ten classes. Fig. 3 shows a sample image of each type of this dataset.

5.2 Model Parameters and Performance Metrics

The most crucial task is finding the best hyper-parameters value to fit the model. The hyper-parameter optimization framework available for machine learning and black-box optimization solvers is known as Optuna (3). Optuna is a

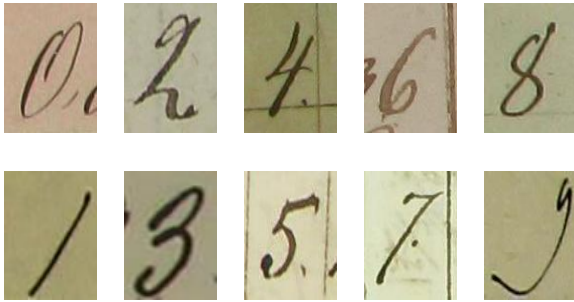


Fig. 3 Samples of digits in DIDA dataset

black-box optimizer, which means that it takes an objective function that provides a numerical value to evaluate the hyperparameters' behavior and select where to sample in future trials. The trial object of optuna defines the kind and range of hyper-parameter to be tuned. The optimizer is chosen from ["Adam", "RMSprop", "SGD"]. The learning rate values vary logarithmically from 0.00001 to 0.1. The batch size is ranged from 16 to 256, with a step size of 16.

The application of Optuna on the EMNIST dataset proposed the best optimizer, stochastic gradient descent (SGD), with a learning rate of 0.01 and batch size of 192. The best dropout value was 0.25, from 0.1 to 1.0. Hence the model is trained using an SGD optimizer with a 0.01 learning rate, 0.9 momentum, and 0.25 dropout. Experiments are done on Google Colab Pro using the PyTorch library. The log_softmax activation function is used for classification. It increases the neural network's interpretability. Loss is computed using negative log-likelihood (NLL) loss.

The model's performance cannot be calculated just by measuring the accuracy of the prediction. Hence in this paper, we evaluate forecasts based on different performance metrics. The most common performance metric is accuracy, calculated by dividing the correct predictions by the total samples.

The other metric, precision, is the number of correct predictions. Recall, also known as sensitivity, is the number of positives returned by the model. Specificity is the opposite of recall, defined as the number of negatives produced by the model. To compute the harmonic mean of precision and recall, f1-score is used. It is a score that represents both precision and recall. The capability of the model in distinguishing classes can be plotted

using Area Under Curve (AUC)-Receiver Operating Characteristic (ROC) metric. These metrics are based on varying threshold values, where AUC measures separability and ROC is a probability curve. Model is termed to be better if AUC is high. AUC-ROC can be created by plotting sensitivity vs. specificity at different threshold values.

5.3 Results

The experimentation uses K-fold cross-validation with ten folds to avoid overfitting as the data is limited. The average tenth fold accuracy achieved on the EMNIST-digit dataset is 99.89%. The comparative performance of the proposed approach with other architectures is shown in Table 1. The

Table 1 Proposed approach accuracy compared to other architectures on EMNIST-digit dataset.

Method	Year	Accuracy (%)
OPIUM (15)	2017	95.90
EDEN (18)	2017	99.30
OptConv+Log+Perc (34)	2020	99.43
CNN (12)	2018	99.46
Parallelized CNN (48)	2017	99.62
Neuro Evolved CNN (20)	2019	99.73
Markov random field CNN (36)	2017	99.75
WaveMixLite-112/16 (24)	2022	99.77
Deep CNN (44)	2018	99.79
TextCaps (23)	2019	99.79
Proposed method	2022	99.89

proposed approach achieved higher accuracy than the state-of-the-art methods. This means vision transformers can be used for handwriting recognition tasks. The precision, recall, and f1-score are the same as accuracy, i.e., 99.89%. The maximum accuracy achieved is 99.95% in the last fold. The roc_auc score of the model is 99.99%. Fig. 4 shows the output on the EMNIST dataset. It displays the actual, predicted value and image of a few samples from the EMNIST-digit dataset. Figure shows that the model displays correct prediction even for rotated images. The confusion matrix of the testing result on the last fold is shown in Fig. 5. The number in the blue color box shows the class-wise correct predictions.

The experimentation is also done on the DIDA dataset to know the effect of the model on original images. The average accuracy, precision, recall,

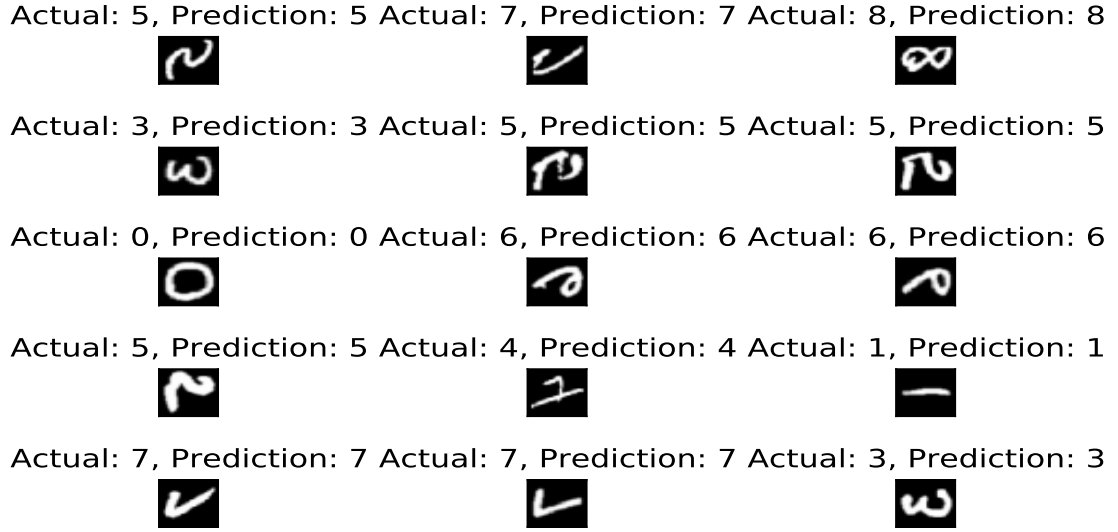


Fig. 4 Recognition result of handwritten digits from EMNIST-digit dataset.

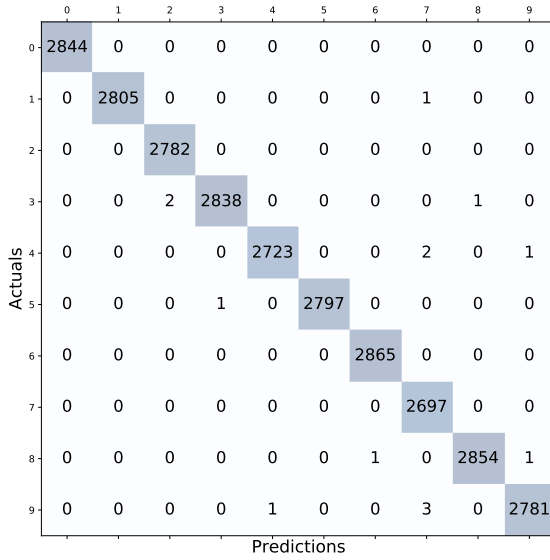


Fig. 5 Confusion matrix for the optimal parameter values examined on EMNIST-digit dataset.

and f1-score are the same, i.e., 99.77%. The confusion matrix supports the computation of sensitivity, specificity, and ROC AUC score. The confusion matrix for maximum accuracy on the

DIDA dataset is shown in Fig. 6. The roc_auc

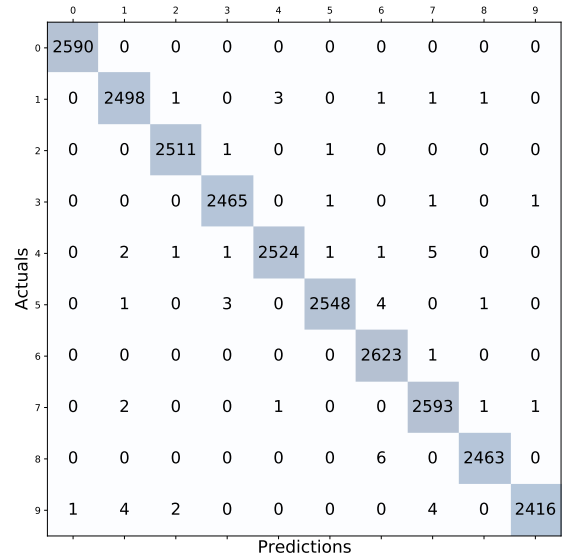


Fig. 6 DIDA dataset confusion matrix with optimized hyperparameter values.

score achieved is 99.99%. The prediction result on a few sample images is shown in Fig. 7. Hence the

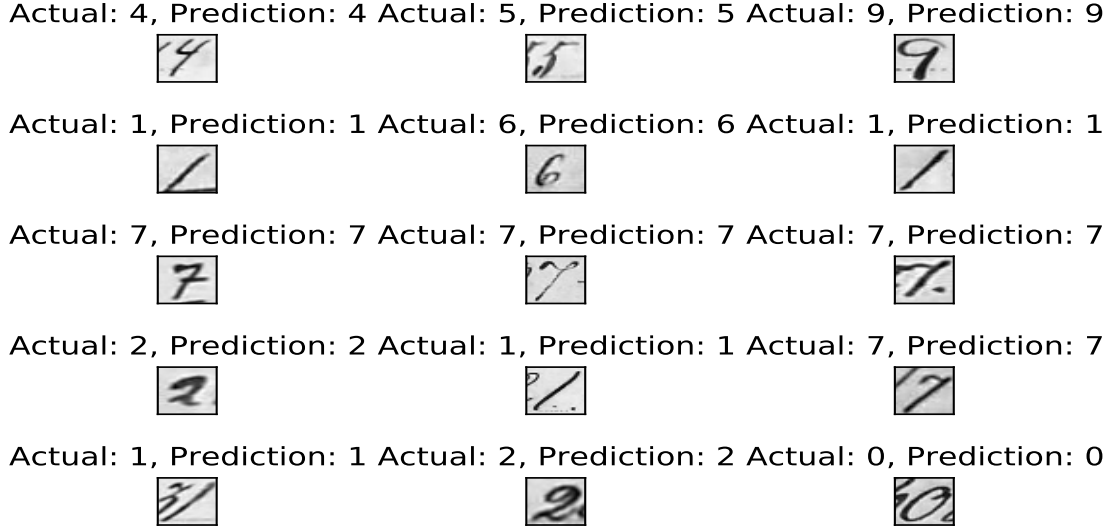


Fig. 7 Prediction on handwritten digits of DIDA dataset.

Table 2 Classwise sensitivity and specificity on EMNIST-digit and DIDA dataset.

Class	EMNIST-digit		DIDA	
	Sensitivity (%)	Specificity (%)	Sensitivity (%)	Specificity (%)
0	100	100	100	99.99
1	99.96	100	99.72	99.96
2	100	99.99	99.92	99.98
3	99.89	99.99	99.88	99.98
4	99.89	99.99	99.56	99.98
5	99.96	100	99.65	99.98
6	100	99.99	99.96	99.95
7	100	99.97	99.81	99.95
8	99.93	99.99	99.76	99.98
9	99.85	99.99	99.55	99.99

model is capable of recognizing digits from original images.

Table 2 shows the class-wise sensitivity and specificity result for both the datasets. The experimentation shows that the model can correctly predict and classify both positive and negative classes.

6 Conclusion

This paper uses a convolutional vision transformer to recognize handwritten digits. The proposed

method is evaluated for two datasets, EMNIST with clean images and DIDA with original photos. The cross-validation approach and tuning of hyper parameters improved the model performance. Three optimizers were used, Adam, SGD, and RMSProp, with varying learning rates, batch size, and dropout. The best recognition accuracy of 99.89% on the EMNIST dataset has been achieved using an SGD optimizer with a 0.01 learning rate, 192 batch size, and 0.25 dropout value. The findings show that the proposed method obtains

better accuracy than other digit recognition techniques. The outcomes demonstrate that the suggested plan is adequate and robust to distortions, writing style, size, direction, and arrangement of images. Hence the model can be used in real-world settings. The testing of the proposed method is done only on a single-digit dataset. The model needs to be extended to recognize digit strings.

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- Conflict of interest
The authors have no competing interests to declare that are relevant to the content of this article.
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Not applicable.
- Consent to participate
Not applicable.
- Consent for publication
Not applicable.
- Availability of data and materials
Not applicable.
- Code availability
Not applicable.
- Authors' contributions
Not applicable.

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